

PA3 — RAG Query Evaluation Report

IEPS-TOLPA | Retrieval-Augmented Generation vs LLM-only

Summary

#	Query	Class	Winner
1	How do I run Segment Anything (SAM) inference?	GOOD	RAG
2	How do I install dependencies for a segmentation projec...	GOOD	RAG
3	What datasets are used for semantic segmentation?	GOOD	RAG
4	How do I download pretrained model weights?	GOOD	RAG
5	What is the weather like in Paris?	BAD	LLM-only
6	How do I bake a chocolate cake?	BAD	LLM-only
7	Who won the 2022 World Cup?	BAD	Neither
8	How do I evaluate segmentation model performance using ...	GOOD	RAG
9	What deep learning frameworks are used for image segmen...	GOOD	RAG

Good queries: 1, 2, 3, 4, 8, 9 (6 total) | Bad queries: 5, 6, 7 (3 total)

Detailed Results

Query 1: How do I run Segment Anything (SAM) inference?

GOOD QUERY

Reasoning: Highly specific technical question directly relevant to the crawled corpus of image segmentation GitHub repositories. RAG retrieved strong, on-topic chunks with positive rerank scores.

RAG mode: Retrieved 5 relevant chunks from MONAI notebooks, Roboflow notebooks, Semantic-SAM, CAT-SAM, and the official Meta SAM repo. Answer provided step-by-step SAM usage guidance grounded in real repo content.

LLM-only: Answered with 'sam -v --help' — a generic CLI guess with no real substance.

Verdict: RAG wins clearly. Retrieved context is directly relevant and the answer is grounded in real repository documentation.

Query 2: How do I install dependencies for a segmentation project?

GOOD QUERY

Reasoning: Practical setup question that appears frequently in GitHub READMEs. RAG retrieved real installation instructions from multiple segmentation repos.

RAG mode: Retrieved chunks from u-Segment3D, android-segmentation, segmentation_training_pipeline, and imgclsmob with concrete pip/conda install commands.

LLM-only: Responded with broken Python code: `'my_segmentation_project = np.array(...)'` — completely nonsensical.

Verdict: RAG wins decisively. LLM-only hallucinated irrelevant code. RAG grounded the answer in real README installation instructions.

Query 3: What datasets are used for semantic segmentation?

GOOD QUERY

Reasoning: Common research question well covered in the corpus. Both modes gave reasonable answers, but RAG provided significantly more detail.

RAG mode: Retrieved chunks listing Cityscapes, PASCAL VOC, COCO, ADE20K, SemanticKITTI, S3DIS, ScanNet, NYUDv2, SUN RGB-D, and Stanford-2D-3D-S with dataset sizes and splits.

LLM-only: Correctly named Cityscapes, COCO, PASCAL VOC, and KITTI, but gave no detail.

Verdict: RAG wins. Both are reasonable, but RAG is richer and more specific, sourced directly from real papers and repos.

Query 4: How do I download pretrained model weights?

GOOD QUERY

Reasoning: Very common README topic across ML repositories. RAG retrieved concrete download instructions from multiple repos.

RAG mode: Retrieved chunks from RAUNE-Net, RMSIN, ISNet, DIS, and SETR with specific download links, directory structure instructions, and file naming conventions.

LLM-only: Hallucinated a fake GitHub repo 'PyTorch/HuggingFaceModels' and a nonexistent 'python -m huggingface models download' command.

Verdict: RAG wins clearly. LLM-only hallucinated. RAG provided real, actionable instructions from actual repos.

Query 5: What is the weather like in Paris?

BAD QUERY

Reasoning: Completely off-topic for a segmentation-focused GitHub corpus. No relevant documents exist in the database.

RAG mode: Retrieved weather-adjacent ML chunks (precipitation forecasting, weather image datasets) with deeply negative rerank scores (worst: -10.75). Despite irrelevant context, model still fabricated a Paris weather answer.

LLM-only: Correctly described Paris climate with seasonal temperature ranges — a factual general knowledge answer.

Verdict: LLM-only wins. RAG was hurt by irrelevant context. This query is entirely outside the corpus scope.

Query 6: How do I bake a chocolate cake?

BAD QUERY

Reasoning: Completely unrelated to the corpus. Included as a stress test for off-topic robustness.

RAG mode: Retrieved COCO dataset training chunks (matched 'COCO' instead of 'cocoa'). All rerank scores negative (worst: -11.17). Model ignored irrelevant context and answered from general knowledge with a vague recipe.

LLM-only: Gave a detailed, accurate chocolate cake recipe with specific quantities and timing.

Verdict: LLM-only wins. RAG retrieved noise and the answer was worse. Notable keyword collision: COCO vs cocoa.

Query 7: Who won the 2022 World Cup?

BAD QUERY

Reasoning: Sports trivia with no connection to the segmentation corpus.

RAG mode: Retrieved 2022 academic paper citations (matched the year '2022'). All rerank scores negative. Model output a literal unfilled template placeholder: '{answer}'.

LLM-only: Answered 'France' — which is factually wrong (Argentina won).

Verdict: Both fail. RAG produced a broken template output. LLM-only hallucinated the wrong winner. Best example of complete failure on both sides.

Query 8: How do I evaluate segmentation model performance using mIoU?

GOOD QUERY

Reasoning: Technical evaluation question directly relevant to the corpus. mIoU is a standard metric mentioned across many segmentation repos.

RAG mode: Retrieved chunks from polyp segmentation, DeepLabV3, road segmentation, and skin segmentation repos — including actual mIoU code snippets using numpy bitwise operations and real benchmark numbers (e.g. 76.42% on Pascal VOC).

LLM-only: Provided a technically incorrect formula involving TN (true negatives), which are not part of the standard IoU calculation. The formula shown was wrong.

Verdict: RAG wins clearly. Retrieved real code and benchmark numbers. LLM-only gave a plausible-sounding but incorrect formula.

Query 9: What deep learning frameworks are used for image segmentation?

GOOD QUERY

Reasoning: Broad but in-domain question well covered across the corpus. Many repos explicitly mention their framework choices.

RAG mode: Retrieved chunks covering U-Net, Attention U-Net, SegNet, DeepLab-LargeFOV in TensorFlow, and PyTorch-based pipelines. Answer correctly identified U-Net, SegNet, and DeepLab as key architectures.

LLM-only: Answered with 'TensorFlow, PyTorch, Keras and OpenCV' — correct but very generic, listing libraries rather than architectures.

Verdict: RAG wins. Provided architecture-level detail (U-Net, SegNet, DeepLab) grounded in real repos, versus a generic library list from LLM-only.

Conclusions

- RAG outperforms LLM-only on all 6 in-corpus queries, providing grounded, specific answers sourced directly from GitHub README content.
- LLM-only outperforms RAG on off-topic queries (weather, baking) where no relevant corpus documents exist — irrelevant retrieved context actively harms the RAG answer.
- Rerank scores are a reliable signal of query relevance: all good queries had positive top scores; all bad queries had deeply negative scores across all chunks.
- Query 8 (mIoU) is a strong example of RAG correcting LLM hallucination — LLM-only gave a technically wrong formula, while RAG retrieved real numpy code and benchmark numbers.
- Query 7 (World Cup) is the most spectacular failure: RAG output a literal '{answer}' placeholder, and LLM-only hallucinated the wrong winner (France instead of Argentina).
- The COCO/cocoa collision in Query 6 highlights a known weakness of embedding-based retrieval: surface-level keyword matches can override semantic intent.
- A confidence threshold on rerank scores could improve robustness — when all scores are negative, the system should indicate it cannot find relevant information rather than answering from noise.